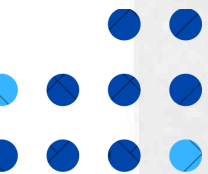


# Constrained Three-Dimensional Container Loading Via Sequential, Embedded, And Repair-Based Intra-Variant Hybrid Grey Wolf Strategies

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**HGWO-MS3D**  
3D Bin Packing Optimizer

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HGWO-MS3D

OR-Library BenchmarkPA

LogisticsResultsVisualizationRun history

CONTAINER (BIN)

Dimensions (cm) — W × D × H

587233220

Max load capacity (kg)

28,000

587 × 233 × 220 cm

28,000 kg max load

ALGORITHM SETTINGS

Hybrid strategy

SequentialEmbedded

Repair-based

Wolf pack sizeMax iterations

30500

Constraints

Fragility (LBS-based)

Allow item rotation

LIFO constraint

7Total items

107.5 kgTotal weight

3Item categories

Item / Box log7 items

Import CSVAdd item

BOX-008W D H 0.00StandardAdd

| ITEM ID | W (CM) | D (CM) | H (CM) | WEIGHT (KG) | QTY | TYPE     |
|---------|--------|--------|--------|-------------|-----|----------|
| BOX-001 | 60     | 40     | 30     | 12.5        | 4   | Standard |
| BOX-002 | 45     | 45     | 45     | 8.0         | 2   | Fragile  |
| BOX-003 | 80     | 60     | 50     | 35.0        | 1   | Heavy    |
| BOX-004 | 55     | 35     | 40     | 18.0        | 3   | Standard |

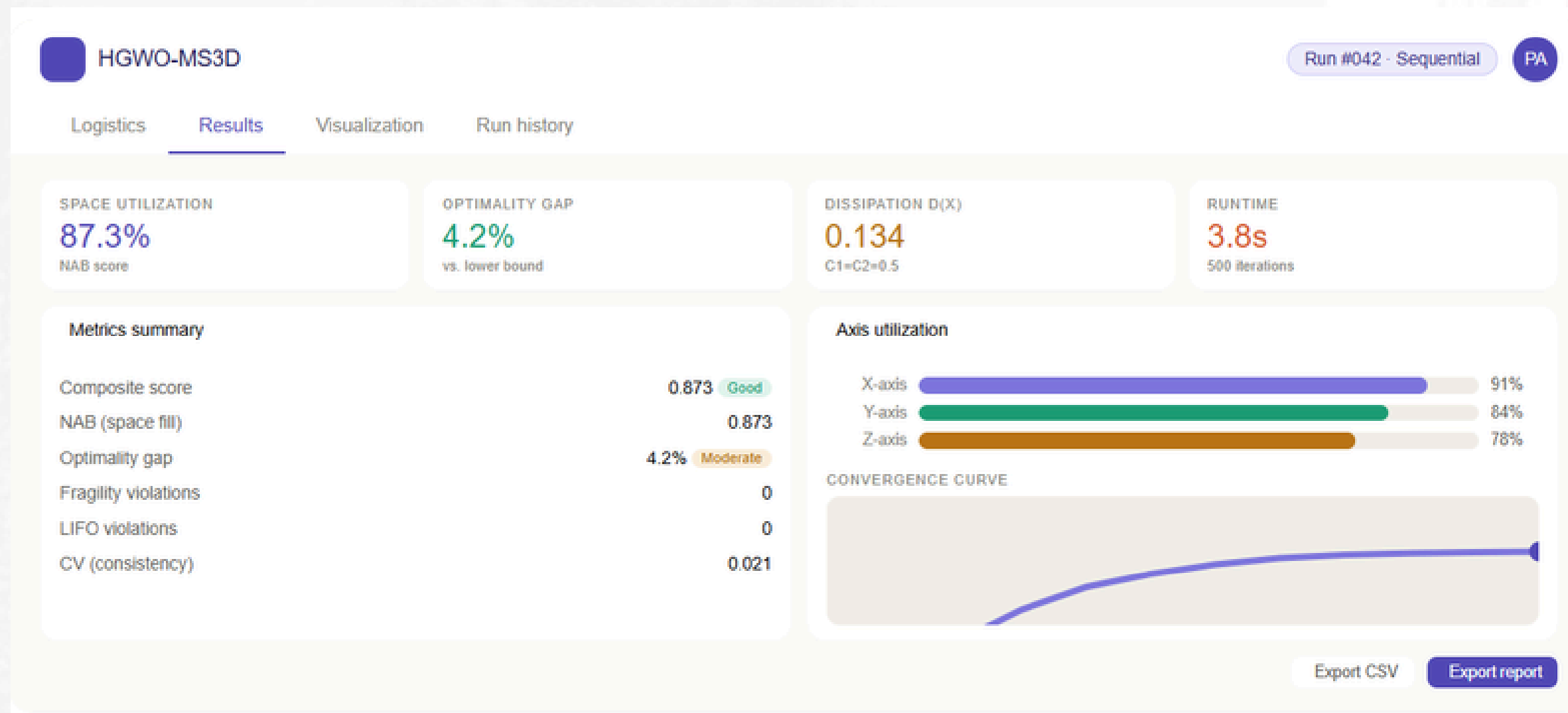
7 items ready

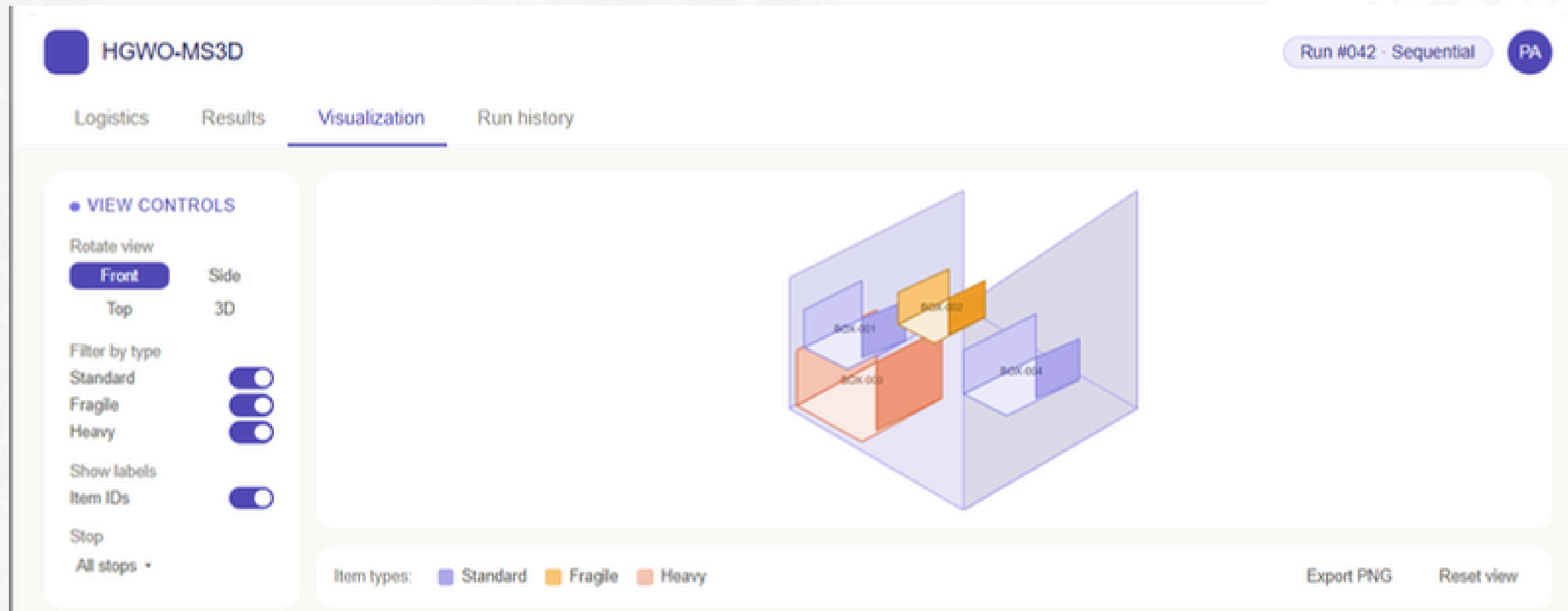
Strategy: Sequential

Total: 107.5 kg

Clear all

Run optimizer









HGWO-MS3D

OR-Library Benchmark

PA

Logistics

Results

Visualization

Run history

Search runs...

All

Sequential

Embedded

Repair-based

Export all



Run #042

Sequential  
Strategy

7 items  
Instance

87.3%  
Space util.

0.134  
D(X)

3.8s  
Runtime

Today 14:02  
Timestamp



Run #041

Embedded  
Strategy

7 items  
Instance

83.1%  
Space util.

0.159  
D(X)

4.1s  
Runtime

Today 13:48  
Timestamp



Run #040

Repair-based  
Strategy

5 items  
Instance

74.6%  
Space util.

0.210  
D(X)

5.3s  
Runtime

Today 12:11  
Timestamp



Run #039

Sequential  
Strategy

10 items  
Instance

91.2%  
Space util.

0.098  
D(X)

6.0s  
Runtime

June 15, 09:30  
Timestamp

